METRIC

MIL-DTL-24728/13

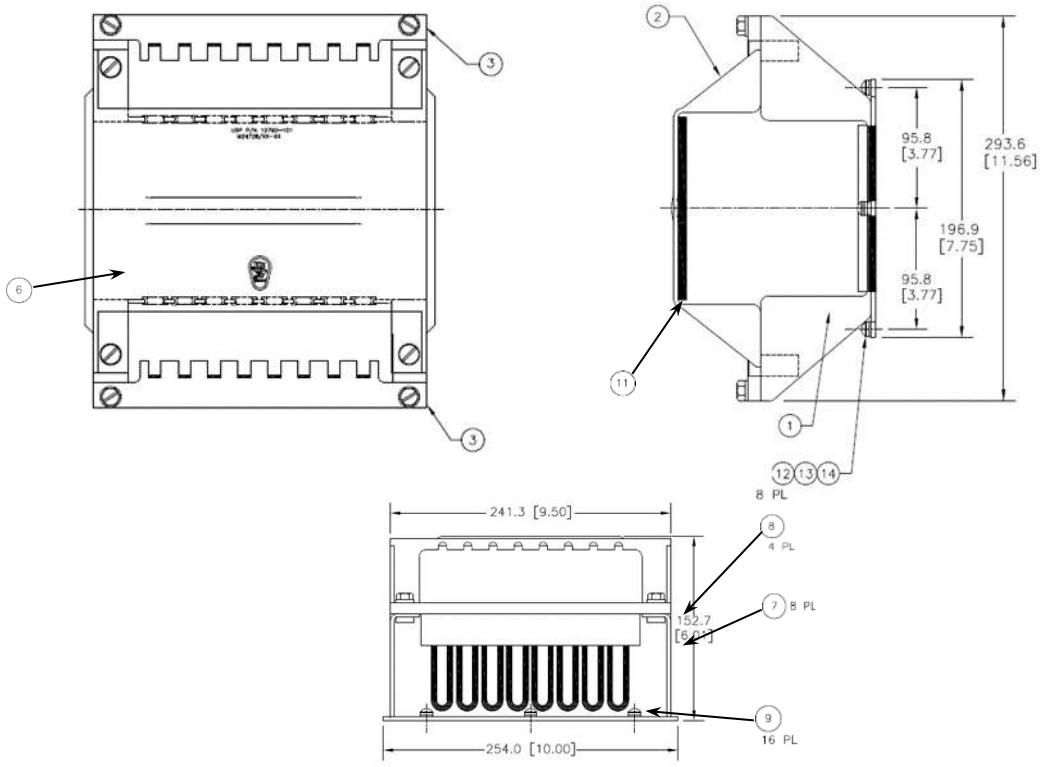
03 September 2019

DETAIL SPECIFICATION SHEET

INTERCONNECTING BOX, FIBER OPTIC, FUSION SPLICE TRAY (24 SPLICE) AND TRAY HOLDER MODULE (8 TRAY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24728.



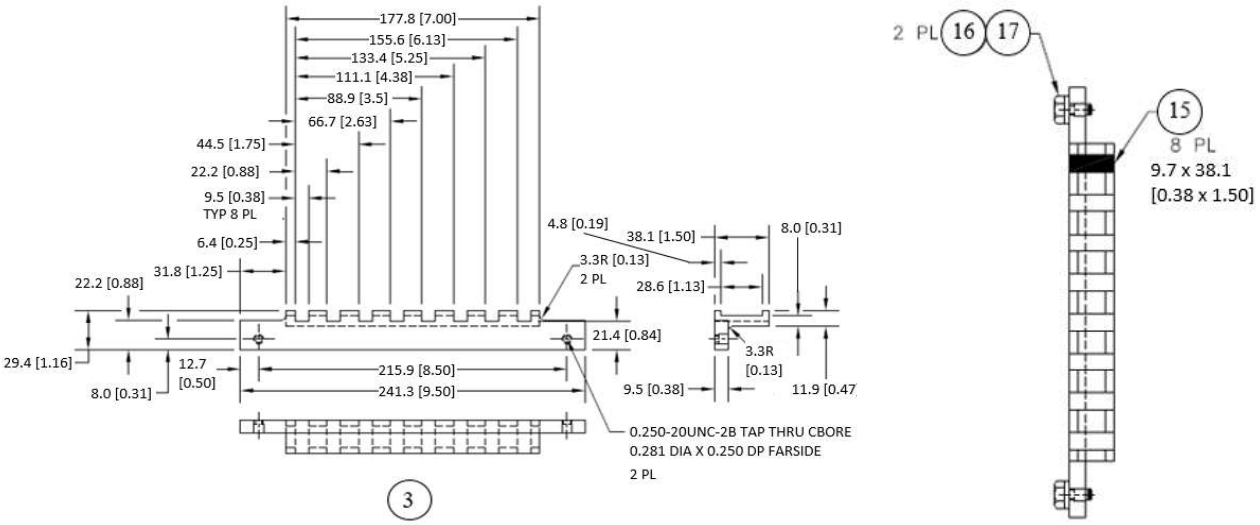
NOTES:

1. For identification of subcomponent parts (circled numbers), see table I.
2. A steel stamp with 3.3-millimeter (0.13-inch) high characters, part or identifying number (PIN), and manufacturer's Commercial and Government Entity (CAGE) code, name, or logo shall be located on subcomponent 2.
3. Base assembly slots (eight places), support angle assembly right slots (eight places), and support angle assembly left slots (eight places) shall be at an angle of 90±1 degrees.
4. The tolerances shall be 0.XX = 0.381 millimeter, 0.XXX = 0.127 millimeter, angles = 0°30' (0.XX = 0.015 inch, 0.XXX = 0.005 inch, angles = 0°30').

FIGURE 1. 10-inch splice tray holder module.

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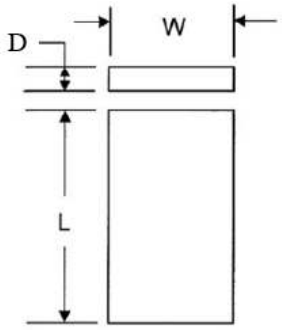
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NOTES:

1. Dimensions shall be as specified in table II.
2. The tolerances shall be 0.XX = 0.381 millimeter, 0.XXX = 0.127 millimeter, angles = 0°30' (0.XX = 0.015 inch, 0.XXX = 0.005 inch, angles = 0°30').

FIGURE 2. Support angle assembly.

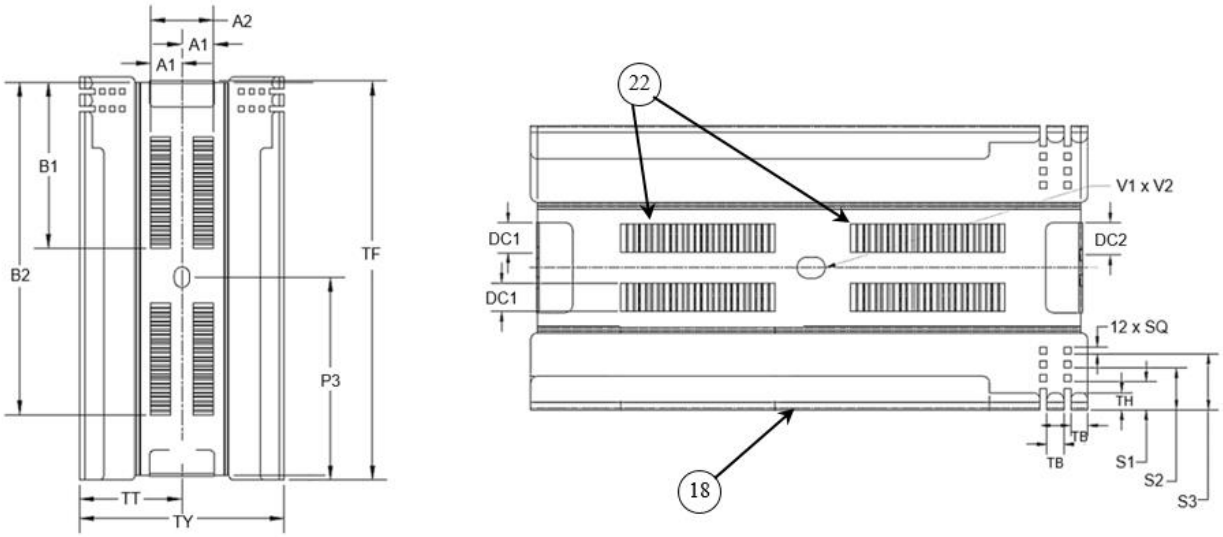


NOTES:

1. Dimensions shall be as specified in table III.
2. Values provided are envelope dimensions.

FIGURE 3. Splice tray envelope (including cover).

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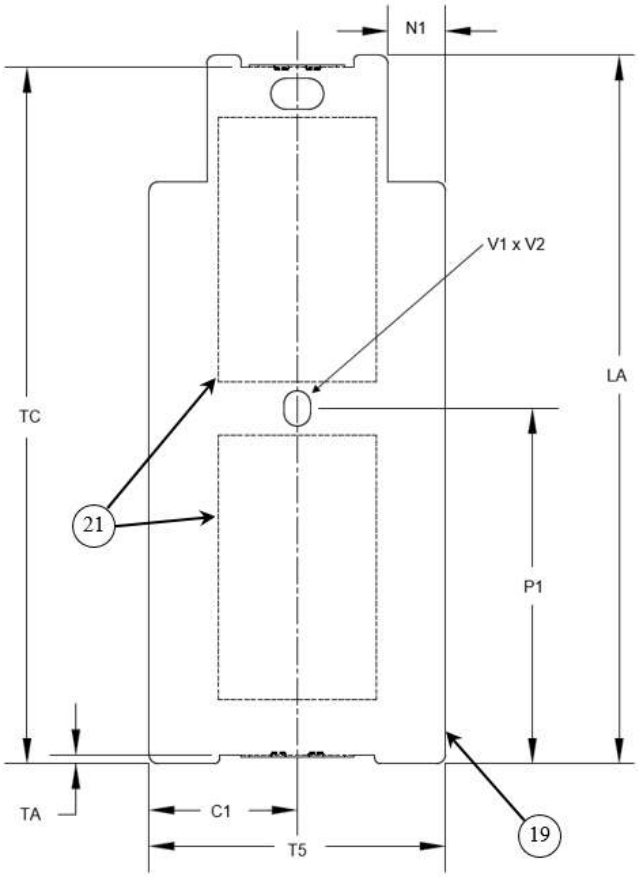


NOTES:

1. Dimensions shall be as specified in table II and table III.
2. Sleeve holders shall be installed such that A1 and A2 dimensions are met.

FIGURE 4. Splice tray base.

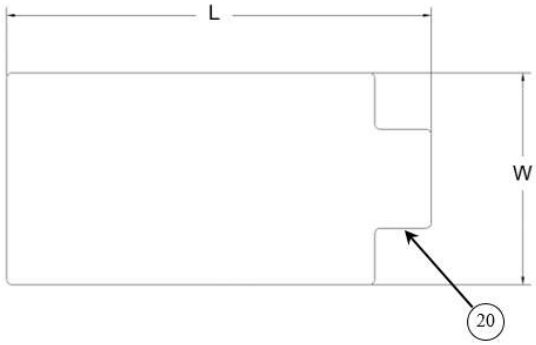
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NOTE: Dimensions shall be as specified in table IV.

FIGURE 5. Splice tray cover.

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NOTE: Dimensions shall be as specified in table V.

FIGURE 6. Mylar insert.

TABLE I. Parts list.

Subcomponent part identification number (circled, figures 1 through 6)	Subcomponent part name	Located in figure(s)	Quantity	Material	Remarks
1	Base assembly	1	1	See remarks	Comprised of subcomponents 5, 7, 9, and 11
2	Cover assembly	1	1	See remarks	Comprised of subcomponents 6, 8, and 11
3	Support angle assembly	1 and 2	2	Aluminum alloy, bar, 6061-T6511	Comprised of subcomponents 15, 16, and 17; 0.50 x 1.50 in; ASTM B221
6	Cover	1	1	Aluminum alloy, 5052-H32	3.175 mm (0.125 in); ASTM B209
7	Clinch nut	1	8	Stainless steel	¼-20UNC-2B, CLS-0420-2
8	Panel screw	1	4	Stainless steel	¼-20UNC-2A
9	Edging grommet	1	16	DTI SL6 0100-ER0201, or equal	---
11	Bumper	1	2	Silicone foam, (P/N) 1/ 8608K524, CAGE OKVE6, or equal	6.35 mm (0.25 in) thick
12	Screw, cross recessed pan head machine	1	8	Stainless steel	¼-20UNC-2A x 0.63LG
13	Washer, flat	1	8	Stainless steel	6.35 mm (¼ in)

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TABLE I. Parts list – Continued.

Subcomponent part identification number(circled, figures 1 through 6)	Subcomponent part name	Located in figure(s)	Quantity	Material	Remarks
14	Washer, split lock	1	8	Stainless steel	6.35 mm (¼ in)
15	Bumper	2	16	Silicone foam, P/N 8608K521, CAGE OKYE6, or equal	1.52 mm (0.06 in) thick
16	Captive bolt	2	4	Stainless steel	Reduced shank; 0.250-20UNC-2A x 0.75LG
17	Flat washer	2	4	Stainless steel	6.35 mm (¼ in)
18	Splice tray base	4	1	Aluminum, 5052	0.76 mm (0.03 in) thick
19	Splice tray cover	5	1	Aluminum, 5052	0.76 mm (0.03 in) thick
20	Mylar insert	6	1	Transparent mylar	0.29 mm (0.011 in) thick, max
21	Cover pad	5	2	Adhesive-backed closed-cell foam	1.6 mm (0.063 in); white
22	Sleeve holder	4	4	Adhesive-backed neoprene rubber	5.56 mm (0.219 in) thick, max; slate gray
NOTE: 1/ Part number (P/N).					

TABLE II. Splice tray base dimensions.

Letter (figure 4)	Characteristic	Millimeters		Inches	
		Min	Max	Min	Max
DC1	Dimple center, bottom	13.21	13.72	0.52	0.54
DC2	Dimple center, top	14.48	14.99	0.57	0.59
P3	Length to oval	124.46	128.27	4.90	5.05
S1	Side to hole one	11.65	13.30	0.46	0.52
S2	Side to hole two	18.35	19.90	0.72	0.78
S3	Side to hole three	24.75	25.95	0.97	1.02
SQ	Hole width	3.05	3.23	0.12	0.13
TB	Tab width	7.49	8.04	0.29	0.32
TF	End to near lip	249.18	250.22	9.81	9.85
TH	Tab extension	7.38	8.08	0.29	0.32
TT	Side to oval center	63.35	64.59	2.49	2.54
TY	Splice tray width	128.30	129.03	5.05	5.08
V1	Oval length	9.39	9.65	0.37	0.38

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TABLE II. Splice tray base dimensions – Continued.

Letter (figure 4)	Characteristic	Millimeters		Inches	
		Min	Max	Min	Max
V2	Oval width	12.48	12.83	0.49	0.51
A1	Center to edge of splice sleeve holder	16.15	25.15	0.64	0.99
A2	Outer to outer edge of splice sleeve holders	39	40.9	1.54	1.61
B1	Top edge to bottom of first splice sleeve holder	98.40	110	3.87	4.33
B2	Top edge to bottom of second splice sleeve holder	202.6	214.2	7.97	8.43
NOTE: Inch equivalents are given for general information only.					

TABLE III. Splice tray envelope dimensions.

Letter (figure 3)	Characteristic	Millimeters		Inches	
		Min	Max	Min	Max
L	Overall length	255	255.5	10.04	10.06
W	Overall width	128.3	129.03	5.05	5.08
D	Overall depth	9.5	9.7	0.37	0.38
NOTE: Inch equivalents are given for general information only.					

TABLE IV. Splice tray cover dimensions.

Letter (figure 5)	Characteristic	Millimeters		Inches	
		Min	Max	Min	Max
C1	Edge to oval center	53.35	53.85	2.10	2.12
LA	Length overall	255.14	255.40	10.04	10.06
N1	Edge to near notch	20.2	21.2	0.80	0.84
P1	Edge to small oval	126.75	129.29	4.99	5.09
T5	Tray cover overall width	106.75	107.19	4.19	4.22
TA	Tab set back	2.41	3.94	0.09	0.16
TC	Tab to notch	250.15	251.59	9.85	9.91
V1	Small oval length	9.39	9.65	0.37	0.38
V2	Small oval width	12.57	12.83	0.49	0.51
NOTE: Inch equivalents are given for general information only.					

TABLE V. Mylar insert dimensions.

Letter (figure 6)	Characteristic	Millimeters		Inches	
		Min	Max	Min	Max
L	Length	227.7	238.76	8.96	9.40
W	Width	118.75	120	4.68	4.72
NOTE: Inch equivalents are given for general information only.					

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Scope. The detail requirements specified herein cover fusion splice trays and tray holders for usage within fiber optic interconnection boxes suitable for military use.

Part or identifying number (PIN). The PIN consists of the letter "M" followed by the base specification number, the specification slant sheet number, and a PIN code (see table VI).

M	**24728**	**/13**	**-**	**XX**
Prefix for Military Specification	Specification Number	Spec Sheet No.		PIN code (see table VI)

TABLE VI. PIN code.

PIN code	Description	Figures
01	Splice tray holder with mounting hardware, for use with 129 x 255 mm (5 x 10 in) trays.	1 and 2
02	Splice tray, 129 x 255 mm (5 x 10 in).	3 through 6

REQUIREMENTS:

Materials. Unless otherwise specified, materials shall be as specified in table I.

Parts and processes. Unless otherwise specified herein, parts and processes shall be selected or developed in accordance with the supplier's standard practices and shall allow the resultant product to meet the requirements of this specification.

Finish, interior and exterior surfaces. After weld and before installing non-metallic components, the surfaces shall be finished either with a coating of epoxy powder in accordance with MIL-PRF-24712 or with a coating of chemical conversion materials in accordance with MIL-DTL-5541, type II, class 1A (aluminum or aluminum alloys only). Refer to MIL-HDBK-2036 for guidance on painted surfaces.

Nonmagnetic materials. Not applicable.

Design and construction. The physical characteristics of the splice tray holder and splice tray shall be as shown on figures 1, 3, and 5 and shall be such that the holder installs into the mounting holes in accordance with MIL-DTL-24728/12 and does not affect the performance of MIL-DTL-24728 fiber optic interconnection boxes, including the ability to properly install the cover. Mounting hardware shall be supplied with each tray holder.

Enclosure. Splice trays and tray holders shall be designed such that all cables, cable components, fibers, connectors, splices, couplers, and mounting and stability supports for the above components entering, enclosed in, or exiting the components are not damaged during servicing of the components or during installation of additional components.

Mounting provisions. The splice tray holder shall install into the mounting holes in accordance with MIL-DTL-24728/12.

Cable interconnection interface. Cable feed shall not degrade the optical performance of fiber optic cables.

Interconnect organization. Splice tray holders and splice trays shall have provisions for organizing fiber splices as specified herein such that fiber splice organizers may be moved into serviceable positions without damage to any component and such that their operational position shall minimize micro- or macro-bends in any cable, optical fiber cable component (OFCC), or fiber. All buffered fibers, OFCCs, blown optical fiber (BOF) tubes, or fiber ribbons shall be routed within the fiber optic interconnection box using tie wrap rails, cable routing posts, or cable management devices to avoid interference with input cables and other splice tray holders.

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Accessibility. All parts of the organizer for fiber interconnecting, splicing, maintenance, mounting, and cable additions shall be accessible from the front. Splices and fiber organizers shall be accessible without the removal of or damage to other splices and fibers.

Mass.

Splice tray. 150 grams (0.33 pound) maximum.

Splice tray holder. 2,404 grams (5.3 pounds) maximum.

Stress relief. If manufacturing induces stress in the material, the splice tray holder shall be stress-relieved in accordance with MIL-E-24142.

Cable retention. Applicable for the splice tray only. Splice trays shall be tested in accordance with TIA-455-6. The test shall be performed with trays mounted in tray holders qualified to this specification and containing optical fiber splices qualified in accordance with MIL-PRF-24623/6. The minimum fiber-to-fiber splice tray pull-out strength shall be 14 newtons (3.1 pounds-force). The fiber shall meet the optical requirements specified herein during and after the test. The pull-out force shall be tested by applying the specified axial tensile load between the OFCC and the splice tray housing for a duration of 1 minute, minimum. The change in optical transmittance shall be monitored during and after the test. The OFCC, fiber, and splices shall show no evidence of jacket damage, clamp failure, distortion from bending of splice parts, or cable disengagement from the clamp.

Cable seal flexing. Not applicable.

Cable twist. Not applicable.

Compression resistance. Not applicable.

Impact resistance. Not applicable.

Operating and storage temperature range. The operating temperature range shall be -28 to 65 °C and the storage temperature range shall be -40 to 70 °C.

Salt spray. Not applicable.

Water pressure. Not applicable.

Fluid immersion. Not applicable.

Structural integrity. The assembled splice tray holder module shall have a resonant frequency not less than the maximum frequency specified in accordance with MIL-STD-167-1, type I.

Flame spread. Not applicable.

Flammability. Not applicable.

Identification markings. The splice tray and tray holder shall be identified individually with markings that are permanent, clearly visible, and legible. Identification marking shall include the PIN and the manufacturer's CAGE code, name, or logo.

Splice tray compatibility. The splice tray shall meet the requirements specified herein and as shown on figures 3 through 6 when used with a qualified splice tray holder, as shown on figures 1 and 2.

Splice tray density. The splice tray shall be able to hold at least 24 splices in accordance with MIL-PRF-24623/6.

Splice tray holder compatibility. The splice tray holder shall meet the requirements specified herein and as shown on figures 1 and 2 when used with a qualified splice tray, as shown on figure 2.

Splice tray holder organization. The splice tray holder shall hold eight MIL-DTL-24728/13-02 splice trays as shown on figures 3 through 6. Tray holders shall be fabricated to provide a minimum of 21.6 millimeters (0.85 inch) distance from tray center line to tray center line when trays are installed.

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VERIFICATION:

Qualification inspection and sample size. Fiber optic splice tray modules and splice trays shall be inspected for qualification as specified in table VII.

Conformance inspection. Conformance inspection shall consist of the inspections and tests specified for groups A, B, and C (table VII).

TABLE VII. Qualification and conformance inspection.

Inspection	Requirement paragraph in MIL-DTL-24728	Verification method paragraph in MIL-DTL-24728	M24728/13-01 Qualification sample size	M24728/13-02 Qualification sample size	M24728/13-01 Conformance sample size	M24728/13-02 Conformance sample size	Conformance inspection group 4/
Group I							
Visual and mechanical	3.4, 3.7	4.6.1	1	3	100% 3/	100% 3/	A
Structural integrity	3.5.3.7.1	4.6.1	1	2	-	-	-
Mass	3.4.8	4.6.3.1	1	3	100% 3/	100% 3/	A
Size	3.4.9	4.6.3.2	1	3	100% 3/	100% 3/	A
Identification markings	3.6	4.6.1	1	3	100% 3/	100% 3/	A
Group II							
Temperature life	3.5.3.1	4.6.6.1	1 2/	2 2/	1	2	C 2/
Thermal shock	3.5.3.2	4.6.6.2	1 2/	2 2/	1	2	C 2/
Operating temperature	3.5.3.10	4.6.6.10	1 2/	2 2/	1	2	C 2/
Temperature/humidity cycling	3.5.3.3	4.6.6.3	1 2/	2 2/	1	2	C 2/
Shock	3.5.3.8	4.6.6.8	1/ 2/	2 2/	1	2	C 2/
Vibration	3.5.3.7	4.6.6.7	1/ 2/	2 2/	1	2	C 2/
Fungus resistance	3.5.4.1	4.6.7.1	Polymeric only	Polymeric only	-	-	-
Cable retention	3.5.2.1	4.6.5.1	-	1	-	2	C 2/

NOTES:

1/ The same samples shall be used in the temperature life inspection.

2/ Qualification testing and conformance group C testing shall be tested while installed in a M24728/12 fiber optic interconnection box. Optically monitored channels shall be fusion spliced using M24623/6 fusion splice sleeves.

3/ Unless alternative sampling plan has been approved (MIL-DTL-24728).

4/ Qualification by similarity considerations may be applied to conformance inspection for group C tests.

Qualification by similarity. Manufacturers who produce product for both MIL-DTL-24728/11 and MIL-DTL-24728/13 who are qualified under M24728/11 and pass the shock and vibration tests and group I inspections as specified herein are qualified under this specification sheet for M24728/13-01. This qualification by similarity is applicable if the same materials and construction are used in the previously qualified MIL-DTL-24728/11 splice tray holder and the splice tray holder under test.

Change in optical transmittance. Applicable with the following modification: The change in optical transmittance during or after any specified environmental or mechanical requirement shall be not greater than the value specified for fusion splices in accordance with MIL-PRF-24623/6.

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Discontinuity. Applicable with the following modification: A discontinuity is a reduction of signal strength for the duration specified for fusion splices in accordance with MIL-PRF-24623/6.

Temperature-life (life aging). Applicable with the following modifications:

a. The specimen shall be exposed to dry air at 85 +3/-0 °C (185 +5/-0 °F) for a period of 240 hours.

b. Optical fiber splices in accordance with MIL-PRF-24623/6 that are stored in the tray and tray holder shall meet the requirements for change in optical transmittance.

Thermal shock. Applicable with the following modifications:

a. Testing shall be performed in accordance with TIA-455-71, test condition C-0.

b. The requirements for change in optical transmittance are not applicable.

Temperature/humidity cycling. Applicable with the following modifications:

a. Testing shall be performed in accordance with TIA/EIA-455-5, method B.

b. Optical fiber splices in accordance with MIL-PRF-24623/6 that are stored in the tray and tray holder shall meet the requirements for change in optical transmittance during and after the test.

Vibration. Applicable with the following modifications:

a. The vibration test shall be performed with optical fiber splices assembled on M85045/16-02 cables in accordance with MIL-PRF-24623/6 that are stored in the tray and tray holder.

b. The optical fiber splices shall meet the requirements for discontinuity during the test and the requirements for change in optical transmittance after the test.

Shock. Applicable with the following modifications:

a. The lightweight shock test shall be performed with optical fiber splices assembled on M85045/16-02 cables in accordance with MIL-PRF-24623/6 that are stored in the splice tray and tray holder.

b. The optical fiber splices shall meet the requirements for discontinuity during the test and the requirements for change in optical transmittance after the test.

Operating temperature (temperature cycling). Applicable with the following modifications:

a. The change in optical transmittance shall be measured during and after the test. Optical fiber splices in accordance with MIL-PRF-24623/6 that are stored in the tray and tray holder shall meet the requirements for change in optical transmittance.

b. A post-test visual examination of the test specimens shall reveal no leakage of waterproofing compounds or other apparent loss of sealing capability, no surface or identification marking impairment, and no damage detrimental to the operation of the test specimens.

INTENDED USE. Fusion splice trays and splice tray holders are intended for use inside protective enclosures. The M24728/13-02 10-inch tray is designed for use with the M24728/13-01 10-inch tray holder. The M24728/13 10-inch tray holder is designed for use in the M24728/12 fiber optic interconnection box.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85

Review activities:
Army – AR, MI
Navy – AS, CG, MC, YD
Air Force – 19
DLA – CC, GS

Civil agency:
NASA – NA

Preparing activity:
Navy − SH
(Project 6099-2019-002)

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